

Quantitative, genome-wide analysis of eukaryotic replication initiation and termination.

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Many fundamental aspects of DNA replication, such as the exact locations where DNA synthesis is initiated and terminated, how frequently origins are used, and how fork progression is influenced by transcription, are poorly understood. Via the deep sequencing of Okazaki fragments, we comprehensively document replication fork directionality throughout the *S. cerevisiae* genome, which permits the systematic analysis of initiation, origin efficiency, fork progression, and termination. We show that leading-strand initiation preferentially occurs within a nucleosome-free region at replication origins. Using a strain in which late origins can be induced to fire early, we show that replication termination is a largely passive phenomenon that does not rely on cis-acting sequences or replication fork pausing. The replication profile is predominantly determined by the kinetics of origin firing, allowing us to reconstruct chromosome-wide timing profiles from an asynchronous culture.